Home Assignment

Week 1 – Linux Shell & Python Fundamentals

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Course Title:
AI/ML

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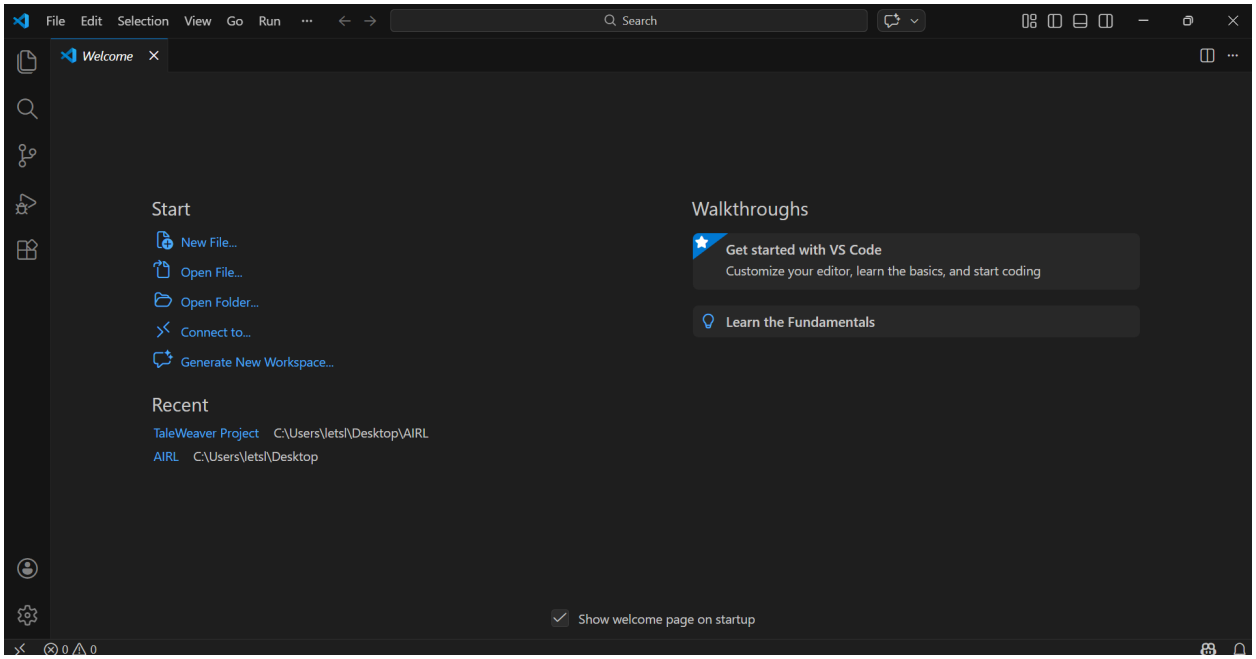
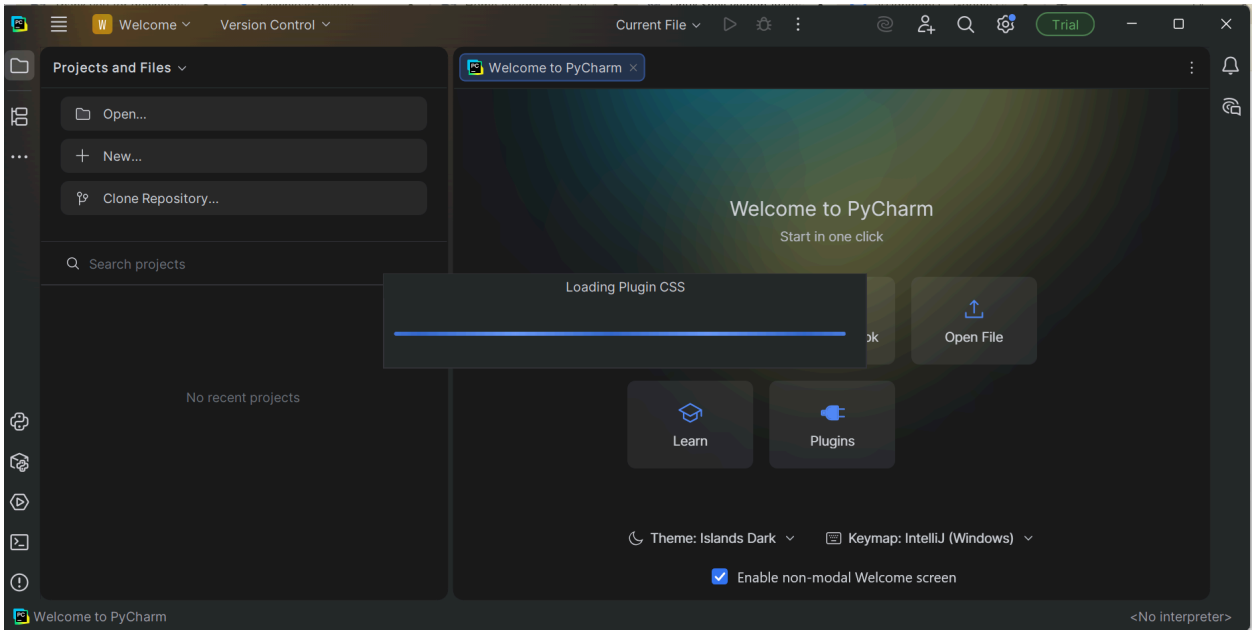
PART 1: SOFTWARE INSTALLATION

Python version installed on the system and Conda version for package management

Visual Studio Code successfully installed and running



PyCharm IDE successfully installed and running



TASK 2: BASIC COMMANDS

Linux Shell commands explanation:

mkdir week1

cd week1

pwd

Creates a directory, navigates into it, and shows current path

mkdir dir1 dir2 dir3

ls

Creates multiple directories and lists them

cd dir1

touch file1.txt file2.txt file3.txt

ls -l

Creates files and displays detailed file information

pwd

Displays current working directory

cd ..

Changes directory

ls -l

Lists files with details

ls -a

Shows hidden files

cat file1.txt

Displays file content

sudo ls

R**uns command with admin privileges**

man mkdir

Shows manual of mkdir command

mkdir testdir

Creates a new directory

rmdir testdir

Removes empty directory

cp file1.txt copy.txt

Copies file

mv copy.txt moved.txt

Moves or renames file

rm moved.txt

Deletes file

echo "Hello" > file1.txt

Writes data to file (overwrite)

echo "World" >> file1.txt

Appends data to file

cat file1.txt

Displays updated content

touch sample.txt

ls

Creates a text file and lists it

nano sample.txt

Creates and edits file using nano editor

file sample.txt

Shows file type

cat sample.txt

Displays full file content

more sample.txt

Displays content page by page

less sample.txt

Allows scrolling through content

head sample.txt

Shows first 10 lines

head -5 sample.txt

Shows first 5 lines

alias showfile="cat sample.txt"

Creates shortcut command

showfile

Executes alias command

shutdown

Displays shutdown command (WSL cannot execute)

sudo reboot

Restarts system

ls -l

Shows current file permissions before change

chmod 777 test.txt

ls -l

Changes file permissions and verifies

ps

Displays running processes

kill <PID>

Terminates a running process

dpkg -l

Lists installed packages

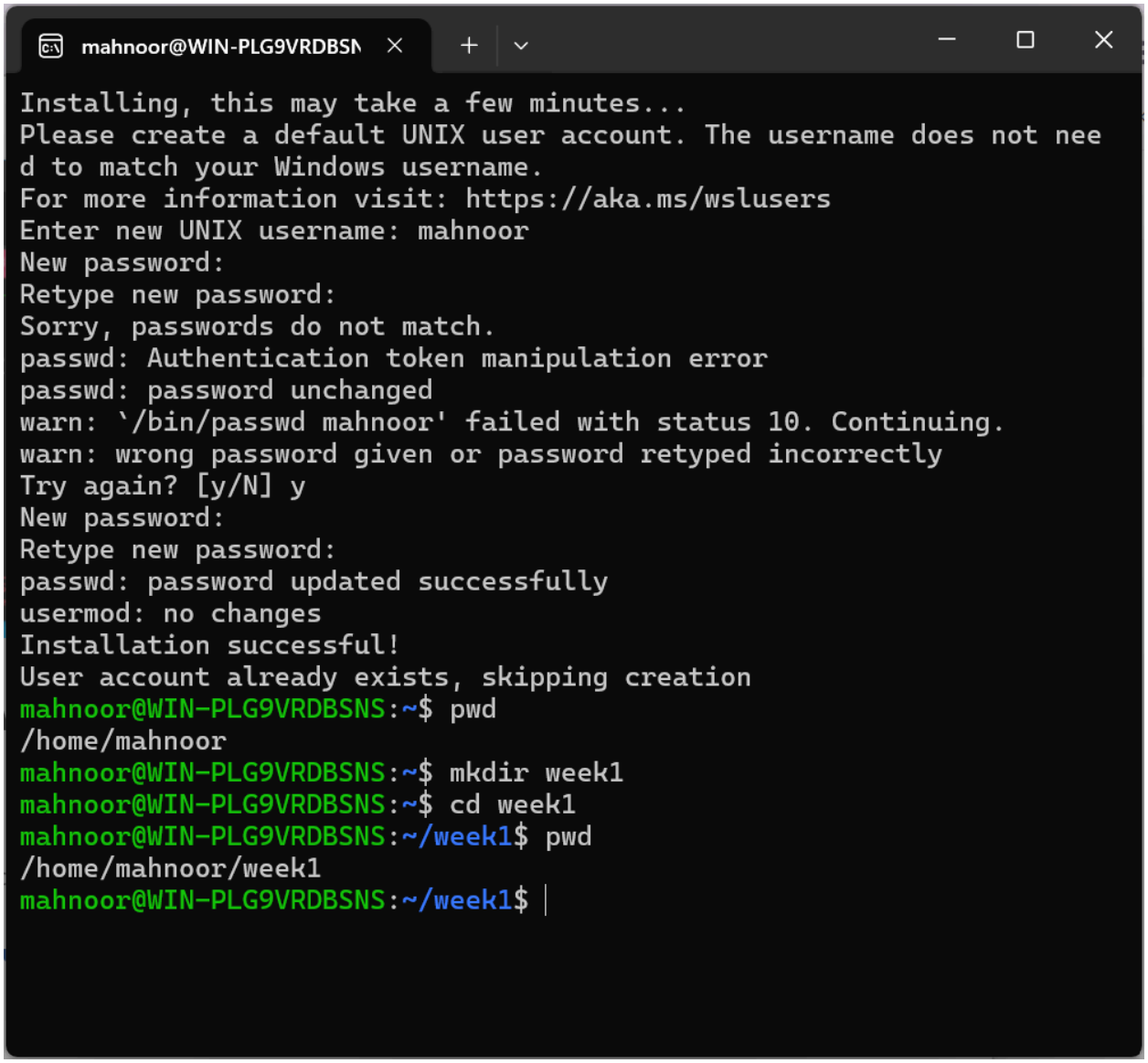
sudo apt update

Updates package list

sudo apt upgrade

Upgrades installed packages

COMMANDS SCREENSHOT



Creates multiple directories and lists them

```
mahnoor@WIN-PLG9VRDBSNS:~/week1$ mkdir dir1 dir2 dir3
mahnoor@WIN-PLG9VRDBSNS:~/week1$ ls
dir1  dir2  dir3
mahnoor@WIN-PLG9VRDBSNS:~/week1$
```

Creates files and displays detailed file information

```
mahnoor@WIN-PLG9VRDBSNS:~/week1$ cd dir1
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ touch file1.txt file2.txt file
3.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls -l
total 0
-rw-r--r-- 1 mahnoor mahnoor 0 Mar  1 11:25 file1.txt
-rw-r--r-- 1 mahnoor mahnoor 0 Mar  1 11:25 file2.txt
-rw-r--r-- 1 mahnoor mahnoor 0 Mar  1 11:25 file3.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

```
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ pwd
/home/mahnoor/week1/dir1
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ cd
mahnoor@WIN-PLG9VRDBSNS:~$ ls -l
total 4
drwxr-xr-x 5 mahnoor mahnoor 4096 Mar  1 11:24 week1
mahnoor@WIN-PLG9VRDBSNS:~$ ls -a
.  ..  .bash_logout  .bashrc  .profile  week1
mahnoor@WIN-PLG9VRDBSNS:~$ cat file
cat: file: No such file or directory
mahnoor@WIN-PLG9VRDBSNS:~$ sudo ls
[sudo] password for mahnoor:
week1
```

```
mahnoor@WIN-PLG9VRDBSN

MKDIR(1)                User Commands                MKDIR(1)

NAME
       mkdir - make directories

SYNOPSIS
       mkdir [OPTION]... DIRECTORY...

DESCRIPTION
       Create the DIRECTORY(ies), if they do not already exist.

       Mandatory  arguments to long options are mandatory for short
       options too.

       -m, --mode=MODE
              set file mode (as in chmod), not a=rwx - umask

       -p, --parents
              no error if  existing,  make  parent  directories  as
              needed,  with  their  file modes unaffected by any -m
              option

       -v, --verbose
              print a message for each created directory

       -Z     set SELinux security context of each  created  direc-
              tory to the default type

       --context[=CTX]
 Manual page mkdir(1) line 1 (press h for help or q to quit)
```

```
mahnoor@WIN-PLG9VRDBSNS:~$ man mkdir
mahnoor@WIN-PLG9VRDBSNS:~$ man mkdir
mahnoor@WIN-PLG9VRDBSNS:~$ mkdir testdir
mahnoor@WIN-PLG9VRDBSNS:~$ rmdir testdir
mahnoor@WIN-PLG9VRDBSNS:~$ cp file1.txt copy.txt
cp: cannot stat 'file1.txt': No such file or directory
mahnoor@WIN-PLG9VRDBSNS:~$ cd week1/dir1
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ cp file1.txt copy.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls
copy.txt  file1.txt  file2.txt  file3.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ cp file1.txt copy.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls
copy.txt  file1.txt  file2.txt  file3.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ^C
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ mv copy.txt moved.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ rm moved.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ echo "Hello" > file1.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ echo "World" >> file1.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ cat file1.txt
Hello
World
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

TASK 3: FILE VIEWING

```
mahnoor@WIN-PLG9VRDBSN
mahnoor@WIN-PLG9VRDBSNS:~$ cd week1/dir1
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ touch sample.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls
file1.txt  file2.txt  file3.txt  sample.txt  sample.txt.save
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ nano sampe.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

```
mahnoor@WIN-PLG9VRDBSN
  GNU nano 8.4                                    sampe.txt
G
LINE 1
INE 2
LINE 3

Write to File: sampe.txt
^G Help          M-D DOS Format     M-A Append         M-B Backup File
^C Cancel        M-M Mac Format     M-P Prepend        ^T Browse
```

```
mahnoor@WIN-PLG9VRDBSN
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ file sampe.txt
sampe.txt: ASCII text
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ cat sampe.txt
G
LINE 1
INE 2
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ more sampe.txt
G
LINE 1
INE 2
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ less sampe.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ head sampe.txt
G
LINE 1
INE 2
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ head -5 sampe.txt
G
LINE 1
INE 2
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

```
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ alias showfile="cat sampe.txt"
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ showfile
G
LINE 1
INE 2
LINE 3

mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

TASK 4: SYSTEM & PROCESS

“Shows the shutdown command, but cannot execute in WSL”

```
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ shutdown
Failed to schedule shutdown: Access denied
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

```
mahnoor@WIN-PLG9VRDBSN
mahnoor@WIN-PLG9VRDBSNS:~$ cd ~/week1/dir1
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ touch test.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls -l
total 12
-rw-r--r-- 1 mahnoor mahnoor 12 Mar  1 11:41 file1.txt
-rw-r--r-- 1 mahnoor mahnoor  0 Mar  1 11:25 file2.txt
-rw-r--r-- 1 mahnoor mahnoor  0 Mar  1 11:25 file3.txt
-rw-r--r-- 1 mahnoor mahnoor 24 Mar  1 11:46 sampe.txt
-rw-r--r-- 1 mahnoor mahnoor  0 Mar  1 11:43 sample.txt
-rw-r--r-- 1 mahnoor mahnoor 31 Mar  1 11:43 sample.txt.save
-rw-r--r-- 1 mahnoor mahnoor  0 Mar  1 11:56 test.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls -l test.txt
-rw-r--r-- 1 mahnoor mahnoor 0 Mar  1 11:56 test.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ chmod 777 test.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ls -l test.txt
-rwxrwxrwx 1 mahnoor mahnoor 0 Mar  1 11:56 test.txt
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

```
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ sleep 1000 &
[2] 257
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ps
    PID TTY          TIME CMD
    238 pts/0    00:00:00 bash
    249 pts/0    00:00:00 sleep
    257 pts/0    00:00:00 sleep
    258 pts/0    00:00:00 ps
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ kill 249
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$ ps
    PID TTY          TIME CMD
    238 pts/0    00:00:00 bash
    257 pts/0    00:00:00 sleep
    259 pts/0    00:00:00 ps
[1]-  Terminated              sleep 1000
mahnoor@WIN-PLG9VRDBSNS:~/week1/dir1$
```

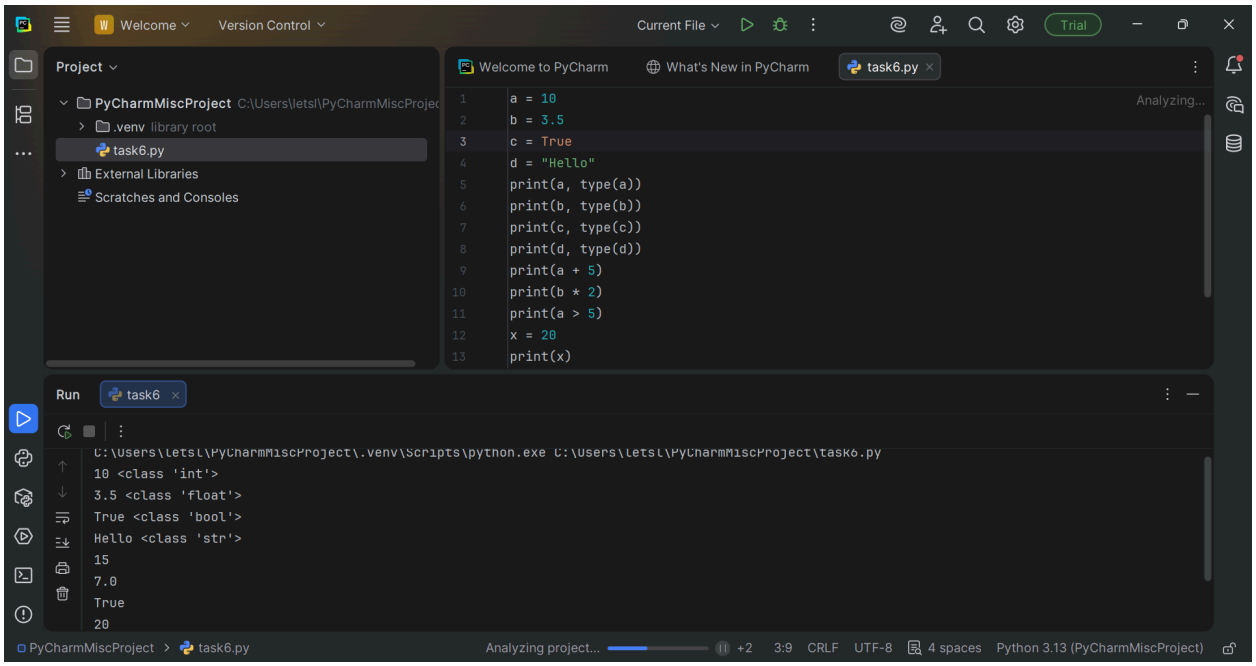
TASK 5: PACKAGE MANAGEMENT

```
mahnoor@WIN-PLG9VRDBSN:~/week1/dir1$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                          Version                      Architecture Description
+++-==========================-============================-============-===============================================>
ii  adduser                       3.152                        all          add and remove users and groups
ii  apt                           3.0.3                        amd64        commandline package manager
ii  apt-utils                     3.0.3                        amd64        package management related utility programs
ii  base-files                    13.8+deb13u2                 amd64        Debian base system miscellaneous files
ii  base-passwd                   3.6.7                        amd64        Debian base system master password and group >
ii  bash                          5.2.37-2+b5                  amd64        GNU Bourne Again SHell
ii  bsdextrautils                 2.41-5                       amd64        extra utilities from 4.4BSD-Lite
ii  bsdutils                      1:2.41-5                     amd64        basic utilities from 4.4BSD-Lite
ii  coreutils                     9.7-3                        amd64        GNU core utilities
ii  cpio                          2.15+dfsg-2                  amd64        GNU cpio -- a program to manage archives of f>
ii  cron                          3.0pl1-197                   amd64        process scheduling daemon
ii  cron-daemon-common            3.0pl1-197                   all          process scheduling daemon's configuration fil>
ii  dash                          0.5.12-12                    amd64        POSIX-compliant shell
ii  dbus                          1.16.2-2                     amd64        simple interprocess messaging system (system >
ii  dbus-bin                      1.16.2-2                     amd64        simple interprocess messaging system (command>
ii  dbus-daemon                   1.16.2-2                     amd64        simple interprocess messaging system (referen>
ii  dbus-session-bus-common       1.16.2-2                     all          simple interprocess messaging system (session>
ii  dbus-system-bus-common        1.16.2-2                     all          simple interprocess messaging system (system >
ii  debconf                       1.5.91                       all          Debian configuration management system
ii  debconf-i18n                  1.5.91                       all          full internationalization support for debconf
ii  debian-archive-keyring        2025.1                       all          OpenPGP archive certificates of the Debian ar>
ii  debianutils                   5.23.2                       amd64        Miscellaneous utilities specific to Debian
ii  dhcpcd-base                   1:10.1.0-11+deb13u1          amd64        DHCPv4 and DHCPv6 dual-stack client (binaries>
ii  diffutils                     1:3.10-4                     amd64        File comparison utilities
```

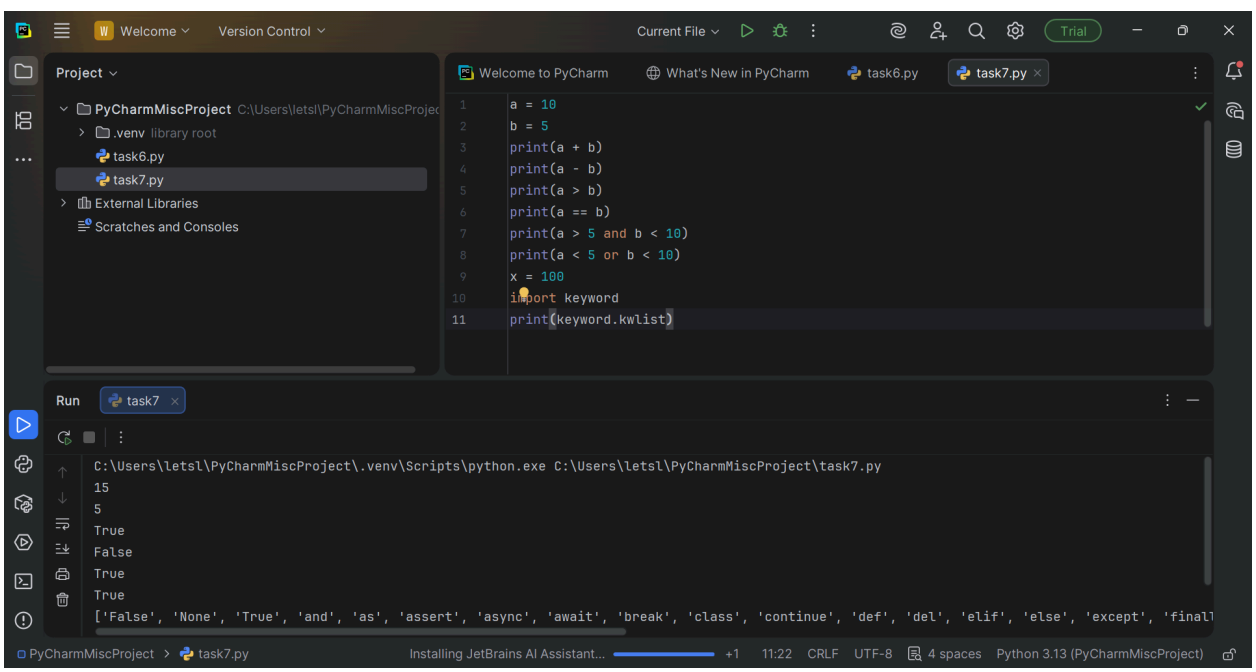
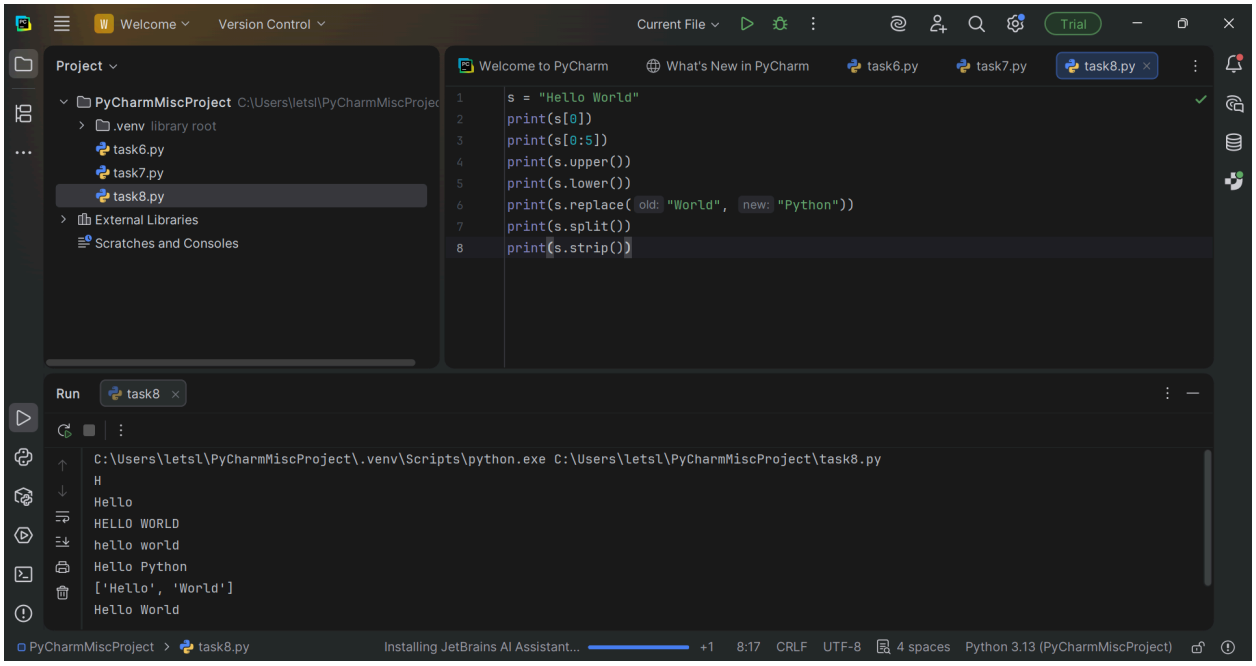
```
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontend), are you root?
mahnoor@WIN-PLG9VRDBSN:~/week1/dir1$ sudo apt update
[sudo] password for mahnoor:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://security.debian.org/debian-security trixie-security InRelease
Hit:3 http://deb.debian.org/debian trixie-updates InRelease
Hit:4 http://ftp.debian.org/debian trixie-backports InRelease
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
mahnoor@WIN-PLG9VRDBSN:~/week1/dir1$ sudo apt upgrade
Upgrading:
  base-files  dhcpcd-base  libc-bin   libc6    libcap2-bin  libext2fs2t64  libssl3t64  logsave                  sqv
  bash        e2fsprogs    libc-l10n  libcap2  libcom-err2  libss2         locales     openssl-provider-legacy

Summary:
  Upgrading: 17, Installing: 0, Removing: 0, Not Upgrading: 0
  Download size: 14.3 MB
  Freed space: 5,120 B

Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libc6 amd64 2.41-12+deb13u1 [2,851 kB]
Get:2 http://security.debian.org/debian-security trixie-security/main amd64 openssl-provider-legacy amd64 3.5.4-1~deb13u
2 [308 kB]
Get:3 http://security.debian.org/debian-security trixie-security/main amd64 libssl3t64 amd64 3.5.4-1~deb13u2 [2,442 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 base-files amd64 13.8+deb13u3 [73.3 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 bash amd64 5.2.37-2+b7 [1,501 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 libc-bin amd64 2.41-12+deb13u1 [637 kB]
Get:7 http://deb.debian.org/debian trixie/main amd64 logsave amd64 1.47.2-3+b7 [24.9 kB]
Get:8 http://deb.debian.org/debian trixie/main amd64 libext2fs2t64 amd64 1.47.2-3+b7 [213 kB]
Get:9 http://deb.debian.org/debian trixie/main amd64 e2fsprogs amd64 1.47.2-3+b7 [591 kB]
Get:10 http://deb.debian.org/debian trixie/main amd64 libcap2 amd64 1:2.75-10+b3 [28.8 kB]
```

PART 3: PYTHON

Task 6: Demonstrates data types, expressions, and statements



Task7:Demonstrates operators, variables, and keywords



Task 8: Demonstrates string operations